HARP: Human-Aware Routing Policies for Multi-Stage Decision Pipelines

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***Abstract*—High-stakes institutional decisions, such as insurance claim review, increasingly combine machine learning with multi-stage human review. Cases are either resolved by a model or escalated through tiers of reviewers with different costs and expertise. Standard selective prediction treats deferral as a binary choice and does not capture these hierarchical pipelines. We introduce Human-Aware Routing Policies (HARP), a framework that treats routing as a learned policy over stages (model vs. reviewer tiers) and minimizes expected system loss under human cost. We formalize single and multi-stage deferral and train a stochastic routing policy with Gumbel-softmax on a loss that combines prediction errors and reviewer costs. We compare HARP to a confidence-threshold naive policy, multi-stage calibrated selective classification (CSC), and an oracle that knows ground truth on Medicare-style data with a two-tier reviewer hierarchy and costs set by marginal risk reduction. Overall, HARP improves the accuracy-cost tradeoff over the naive policy and matches or outperforms CSC while learning routing end-to-end.**

***Keywords*—** Institutional Decisions, Routing, Human-aware

I. Introduction

High-stakes institutional decision like medical insurance claims, immigration applications, and credit approvals are increasingly supported by machine learning systems. In general, these systems are not fully automated decision makers. Instead, automated models are embedded inside structured human review pipelines, where cases can be autonomously resolved or escalated to one or more human reviewers depending on factors including uncertainty, risk, and institutional constraints [1], [2].

This problem is called *selective prediction*, where a model is allowed to abstain on inputs where it has low confidence and defer to an expert human decision-maker [3], [4]. Previous works have studied selective classification and abstention, describing tradeoffs between prediction risk and coverage. Works such as [4], [5] have developed confidence-based deferral policies with strong theoretical guarantees. These approaches have been shown to be effective in applications where the alternative to automated prediction is a single, homogenous human expert.

However, real-world decision pipelines rarely conform to single-stage abstraction. In practice, human review is hierarchical, where cases can be routed to reviewers with differing levels of expertise, authority, and cost. For example, in insurance claim resolution, an application may be resolved automatically, reviewed by a junior examiner, escalated to a senior specialist, or forwarded to an administrative judge. Modelling deferral as a binary decision process does not always model this system well as it obscures important tradeoffs between accuracy and human resource allocation.

Previous work on algorithmic triage and human-AI collaboration has studied how automated systems can allocate cases between humans and models [6], [7], [8]. While these approaches acknowledge heterogenous human expertise, they focus on two-stage settings, rarely on heuristic or fixed routing rules, or do not explicitly connect to the selective prediction framework. As a result, existing methods provide limited guidance for designing multi-stage pipelines under explicit budget constraints.

In this work, we introduce Human-Aware Routing Policies (HARP), a framework for selective prediction in multi-stage human-AI decision pipelines. Rather than making deferral a binary action, HARP formulates decision making as a routing problem over a hierarchy of human reviewers with increasing costs, with the objective of minimizing system loss while also respecting human resource constraints.

We study this framework in the context of structured application review pipelines motivated by insurance claim pipelines, including Medicare and Social Security Disability Insurance claim resolution, where decisions are sequential, resource constrained, and subject to high institutional stakes. This setting shows the limitations of single-stage selective prediction and motivates the need for routing-aware decision policies.

Our work makes the following contributions:

- We formalize multi-stage selective prediction with a hierarchy of human reviewers and extend the single-stage risk–coverage formulation to a budget-constrained routing objective (Eq. (11)).
- We propose HARP: a routing policy π that takes encoder features and model uncertainty (entropy of $f(x)$), trained with Gumbel-softmax on a system loss that combines prediction errors and reviewer costs (Sec. IV).
- We evaluate on CMS-style claim data with a simulated two-tier reviewer hierarchy and MRR reviewer costs. HARP improves over a confidence-threshold naive policy and is competitive with or better than multi-stage CSC on the accuracy–cost tradeoff.

II. Problem Formulation

A. Single Stage Policies

We specifically study human-AI collaborative decision making in structured application review pipelines, where an automated system can either issue a decision or defer the decision to a human reviewer at some cost.

Let

$(x, y) \sim D$, where D is the real world distribution over applications and ground-truth decisions
$x \in \mathcal{X}$, where x is an application
$y \in \mathcal{Y}$, where y is the ground truth decision

We consider a predictive model $f : \mathcal{X} \to \mathcal{Y}$ and a deferral policy $r : \mathcal{X} \to \{0, 1\}$, where

$r(x) = 0$ when the policy chooses to predict autonomously
$r(x) = 1$ when the policy defers to a human for a cost $c_h > 0$.

When $r(x) = 0$, the automated system incurs cost $\mathcal{L}_1$, which is given by

$$\mathcal{L}_1(f) = \ell(f(x), y),$$

where ℓ is by convention 0-1 loss or cross-entropy.

When $r(x) = 1$, the system defers to a human for a human cost $c_h > 0$, assuming that the human produces the correct decision. Therefore, the total system loss can be modeled as the expectation of a function of f and r:

$$\mathcal{L}(f, r) = \mathbb{E}_{(x,y)\sim D}\Big[\ell(f(x), y) \cdot \mathbf{1}\{r(x) = 0\} + c_h \cdot \mathbf{1}\{r(x) = 1\}\Big] \quad (1)$$

In this selective prediction model, when the policy $r(x) = 0$, f predicts autonomously. Over time, as $r(x)$ continues to be 1, c_h will increase. Therefore, we can define a coverage function of r as follows:

$$\text{cov}(r) = \mathbb{E}_{(x,y)\sim D}[\mathbf{1}\{r(x) = 0\}], \quad (2)$$

which describes the fraction for which the policy r chooses for f not to abstain.

When $r(x) = 0$, $f(x)$ is evaluated and we can define the conditional risk as the loss ℓ between $f(x)$ and y for a given $(x, y) \sim D$. More formally, this is:

$$\text{risk}(f, r) = \mathbb{E}_{(x,y)\sim D}[\ell(f(x), y) \mid r(x) = 0], \quad (3)$$

which shows the conditional risk when f autonomously makes the decision.

Based on this coverage function and the conditional risk $\mathcal{L}$, we can define a coverage-constrained optimization problem as follows:

$$\min_{f,r} \mathbb{E}\big[\ell(f(x), y) \mid r(x) = 0\big], \qquad \text{s.t. } \text{cov}(r) \geq \kappa, \quad (4)$$

where κ is the constraint for the fraction where $r(x) = 0$.

Therefore, we can apply a confidence-based threshold defined on the confidence from the output of f, which is traditionally just the max of the output of the sigmoid or softmax function if f is a neural network. Thus, we let:

$$p_\theta(y|x) \quad \text{be the predictive distribution of } f \quad (5)$$

$$\text{conf}(x) = \max_y p_\theta(y|x) \quad \text{be defined as the confidence of a given application-decision pair.} \quad (6)$$

We can then threshold the policy r on the confidence function for a hyperparameter τ as follows:

$$r_\tau(x) = \begin{cases} 0 & \text{if conf}(x) \geq \tau, \\ 1 & \text{if conf}(x) < \tau \end{cases} \quad (7)$$

B. Extension to Multi-Stage Policies

The optimization problem in Eq. (4) is applicable when we have a binary decision of using policy f or abstaining and deferring to an arbitrary human reviewer. However, when we have a hierarchical set of k human reviewers $H = \{0, 1, \ldots, k\}$, each with increasing costs c_{h_k}, we have c_{h_k} increasing when $k \geq 1$.

For the purposes of this paper, single-stage refers to $|H| = 1$, and multi-stage is when $|H| > 1$. We extend the single stage-policy to multi-stage. In this case, we make a trivial change to the minimization problem in Eq. (4) and the initial routing policy r. Let

$$r_m : \mathcal{X} \to \{0, 1, \ldots, k\} \quad \text{be the multistage routing policy,} \quad (8)$$

$$\ell_s(x) \quad \text{be the loss of sending } x \text{ to stage } s, \quad (9)$$

$$B \quad \text{be the total budget constraint for the human resources available.} \quad (10)$$

We assume $\ell_0(x) = \ell(f(x), y)$ corresponds to the automated model loss, and $\ell_s(x) = c_{h_s}$ for $s \geq 1$, assuming human reviewers produce correct decisions.

Therefore, the constrained optimization problem we must solve is as follows:

$$\min_{r_m} \mathbb{E}\big[\ell_{r_m(x)}(x)\big], \qquad \text{s.t. } \mathbb{E}\big[c_{h_{r_m(x)}}\big] \leq B. \quad (11)$$

C. Stochastic Routing Policies

In practice, stochastic routing allows for calculating gradients in optimization, where the system learns a probabilistic policy over decision stages. Rather than deterministically assigning an application x to a single stage, we define a routing distribution $\pi_\phi(s \mid x)$ over stages $s \in \{0, 1, \ldots, k\}$, parameterized by ϕ. For each input, a stage is sampled according to this distribution, allowing the routing policy to express uncertainty and to smooth decisions near stage boundaries.

Under this formulation, the expected system loss is given by

$$\mathbb{E}_{(x,y)\sim D}\left[\sum_{s=0}^{k}\pi_\phi(s\mid x)\,\ell_s(x)\right], \tag{12}$$

with an associated expected human cost

$$\mathbb{E}_{x\sim D}\left[\sum_{s=0}^{k}\pi_\phi(s\mid x)\,c_{h_s}\right]. \tag{13}$$

Therefore, at training time the system is optimized stochastically, but during inference it is deterministic.

III. Background and Related Work

We overview selective classification and prediction policies, algorithmic triage, and organizational resource management models in current literature.

A. Selective Prediction and The Reject Option

The basis for HARP is selective prediction, where a classifier has a reject option to abstain from making a prediction when a defined uncertainty score is high [3]. These works generally deal with the statistical modelling of the trade-off between risk, which is the error rate, and coverage, the fraction of samples classified by the autonomous model. While previous literature has established confidence-based thresholds with statistical guarantees [4], they typically treat abstention as the end-state of the system. However, in institutional pipelines abstaining is not always the end of the decision, with several hierarchies of human reviewers analyzing samples for different costs.

B. Calibrated Selective Classification (CSC) as a Baseline

One of the main challenges in the model routing domain is ensuring that the error probability of a routing model is accurately captured by model confidence scores. Work by [5], [9] in Calibrated Selective Classification (CSC) has developed methods to provide statistical guarantees on risk by calibrating confidence scores within a separate, non-routing model.

While existing works in CSC typically focus on the binary ”predict” or ”defer” case, we extend these statistical guarantees to serve as a multi-stage baseline. Specifically, by calibrating a sequence of thresholds to partition a confidence space into n distinct regions, where n represents the number of reviewers in the institutional hierarchy, we create a strong calibration-based benchmark to compare with HARP's learned routing policies.

C. Learning to Defer (L2D)

Learning to Defer, or L2D, is a deferral policy where the automated decision-making model and the deferral policy should be optimized together to account for when there exists a human partner [10]. L2D generally assumes a single-stage system with a homogenous pool of experts [7] or just one expert. Our work extends L2D by recognizing that human expertise can be structured hierarchically, especially when cases are escalated to increasingly accurate human reviewer and higher costs are incurred on the institution.

IV. Data Processing

To evaluate the HARP framework, we collect raw administrative medical records from the Centers for Medicare and Medicaid Service Medicare Claims Synthetic Public Use Files (CMS DE1.0 SynPUF 2008-2010). We develop a data pipeline to structure these raw claims in a multi-stage decision environment.

A. Assigning Reviewer Hierarchies

In order to assign learnable reviewer hierarchies, we treat a reviewer n to have a higher complexity than the previous reviewer $n-1$, where a complexity score $S(x)$, for each claim x, is defined as:

$$S(x) = \sum_{i\in\text{indicators}} w_i \cdot 1_i(x), \tag{14}$$

where w_i is the weight for the respective indicator and for the indicator function $1_i(x)$ active when claim x meets any indicator below:

- multi-payer involvement
- a short-stay duration of less than one day
- any completed surgical operation
- a high-cost bill defined as $>$\$20,000

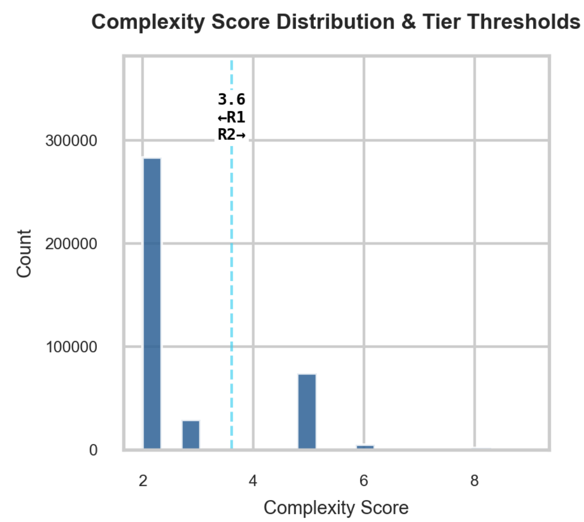

Fig. 1. K-means determined thresholds between $n = 2$ reviewer classes defined across the complexity scores

To partition the complexity scores for each claim into a hierarchy of n reviewers, we apply K-means clustering to the distribution of $S(X)$. Let $B = \{b_1, b_2, \ldots, b_n\}$ be the set of K-means centroids, then we calculate the threshold complexity scores between each reviewer tier $C = \{c_1, c_2, \ldots, c_{n-1}\}$ as the midpoint between the nodes in B, i.e.

$$c_n = \frac{b_n + b_{n+1}}{2}. \tag{15}$$

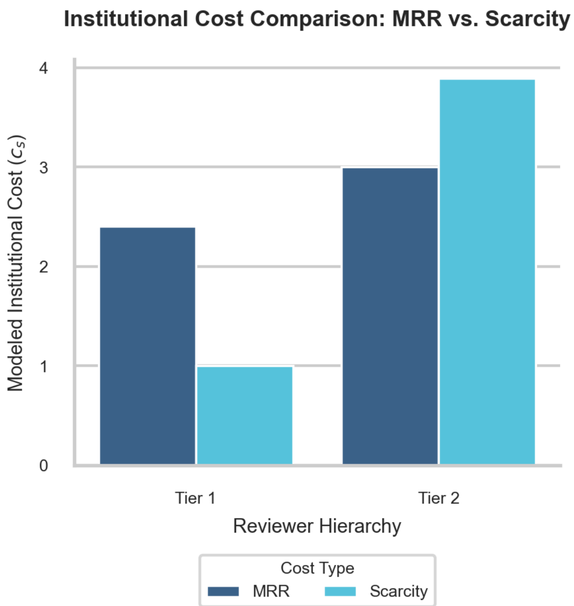

Fig. 2. MRR costs vs. Scarcity costs across reviewer hierarchy show that MRR is best used in this dataset.

B. Determining Reviewer Performance and Costs

With a real-world government dataset containing whether the nth reviewer escalates to the $n+1$th stage or makes a final decision, we could statistically determine the costs of a specific reviewer. We model this by leveraging the claim complexity score $S(X)$ and the reviewer's capabilities.

We predefine error rates $E = \{E_1, E_2, \ldots, E_n\}$ for each reviewer tier, where reviewer $n = \|E\|$ has a 5% error rate, since they must have the "final say" in the claim process and make a small percentage of administrative errors that were not caught at stage $i < n$.

We calculate costs for each reviewer stage up till n reviewer stages as c_s using two distinct methods:

1) Marginal Risk Reduction (MRR): is proportional to expected accuracy gain from escalating to a higher-tier reviewer. If E_t is the error rate of reviewer tier $t < n$, then the cost of a reviewer at tier t c_t is defined as:

$$c_s = \lambda \cdot (E_{t-1} - E_t), \tag{16}$$

for a scaling factor λ.

2) Scarcity: is inversely proportional to the frequency of a specific reviewer tier, so c_t is defined as:

$$c_t = c_{base} \cdot \frac{N_1}{N_t}, \tag{17}$$

where c_{base} is the base cost of reviewer 1.

We use MRR for calculating c_t because MRR can help HARP learn tangible accuracy gains in its cost, as shown by the increasing costs of MRR in Fig. 2.

V. Model Architecture

As shown in Fig. 3, HARP consists of three main models: a shared encoder, f, and π. f is any automated model that outputs accept/deny logits. First, f and the encoder are trained as a warm-start. Once trained, f and the encoder are frozen. π is the routing model HARP learns that takes the output of the frozen encoder and the entropy of f's output logit. Let f output an accept logit for an encoded claim x, then the entropy of $f(x)$ is defined as the Binary Cross Entropy of $f(x)$.

A. Routing and Loss Formulation

As shown in Fig. 3, HARP leverages the gumbel-softmax trick on $\pi(x)$ for a differentiable, "discrete" routing choice. We use this to formulate a total system loss on which we optimize to train π.

We define the total system loss as

$$\mathcal{L}_{system} = r \cdot 1_g(0) + \sum_{i=1}^{n}((1_g(i) \cdot (1 - d_i)) \cdot \lambda_{\text{penalty}}) + c_i, \tag{18}$$

where r is the binary cross entropy of $f(x)$, 1_g is an indicator function that is active if the gumbel softmax of $\pi(x)$ is 1 at index i, d_i is if the reviewer at tier i accurately identified the claim (0 or 1), λ_{penalty} is an error penalty hyperparameter, and c_i is the institutional cost of a reviewer i calculated by the methods described in Sec. IV-B.

We minimize this cost during training and during inference, we route by calculating $\text{argmax}(\pi(x))$ instead of the gumbel-softmax.

VI. Results

We evaluate HARP on the encoded CMS claim dataset with a train/test/validation split, a frozen f, and a reviewer hierarchy with $n = 2$. Reviewer costs c_i are set via Marginal Risk Reduction. We compare (1) Naive: routing by a single confidence threshold on f; (2) Oracle: optimal routing *knowing* ground-truth outcomes (lower bound); (3) Calibrated Selective Classification (CSC): multi-stage calibrated selective with coverage-based threshold; (4) HARP: Learned router π trained with system loss in Eq. 18.

TABLE I
Test-set accuracy and risk by method.

Method	Accuracy	Risk
Naive	0.7999	0.0034
Oracle	**0.9519**	**0.0509**
CSC	0.8754	0.0539
HARP (ours)	**0.8965**	**0.1034**

Table I reports the test-set metrics and mean cost per claim for each method at comparable coverage levels. HARP improves over Naive and outperforms CSC, while performing lower than the Oracle.

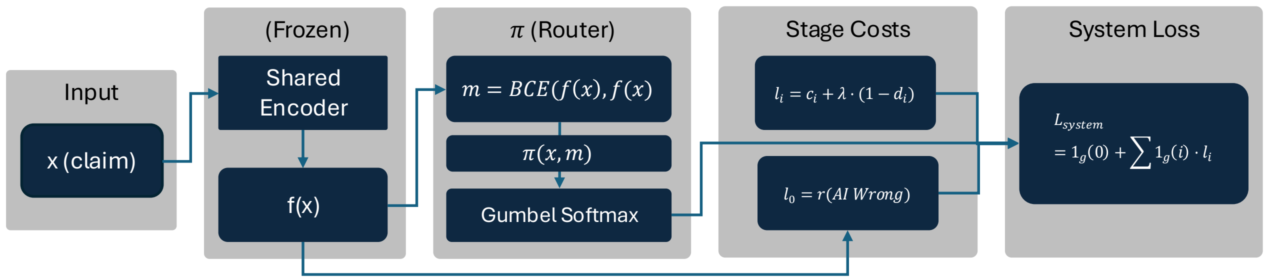

Fig. 3. Human-Aware Routing Policies (HARP) model architecture. We use a shared encoder and frozen f in order to train a router π to learn the HARP routing policy.

VII. Discussion

HARP improves over a single confidence threshold (Naive) by explicitly modeling multiple reviewer tiers and their costs, and performs on par with or better than multi-stage CSC while learning routing end-to-end. The biggest gains are when reviewer costs and error rates differ across tiers. Limitations are as follows: (1) reviewer correctness d_i is simulated from complexity scores rather than observed; (2) costs are set by MRR/scarcity heuristics rather than real institutional data; (3) we do not yet provide formal risk-coverage guarantees as in CSC. These results support learned routing policies in multi-stage pipelines when cost and accuracy tradeoffs are explicit.

VIII. Conclusion and Future Work

We introduced HARP, a framework for human-aware routing in multi-stage decision pipelines. By formulating routing as a stochastic policy $\pi_\phi(s \mid x)$ and minimizing expected system loss (prediction loss plus reviewer costs), HARP learns when to use the model versus each reviewer tier. On CMS-style claim data with two reviewer tiers, HARP outperforms a naive confidence policy and is competitive with or better than multi-stage CSC.

Author Note

The authors declare no competing interests.

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